# IAD-R1: Reinforcing Consistent Reasoning in Industrial Anomaly Detection (AAAI-26)

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2023313030 박민선

# Introduction

## Problem Setting

- Diverse anomaly types + large inter-class variance + scarce anomaly samples → hard to generalization

## Limitations

- Traditional: handcrafted feature + expert knowledge → generalization 부족
- VLM 도입: 멀티모달 이해 + 일반화 능력
  - 기존 AD 결과 + text prompt → 기존 AD 성능에 upper bound 걸림
  - VLM을 직접 fine-tuning / RL로 학습

  → reasoning 없이 input-output mapping만 학습, reward 설계 거칠어서 reasoning–answer 불일치 발생

## Insight & Solution

- CoT + RL → reasoning 성능 향상
- IAD-R1 (reasoning-aware post-training framework)

# Introduction

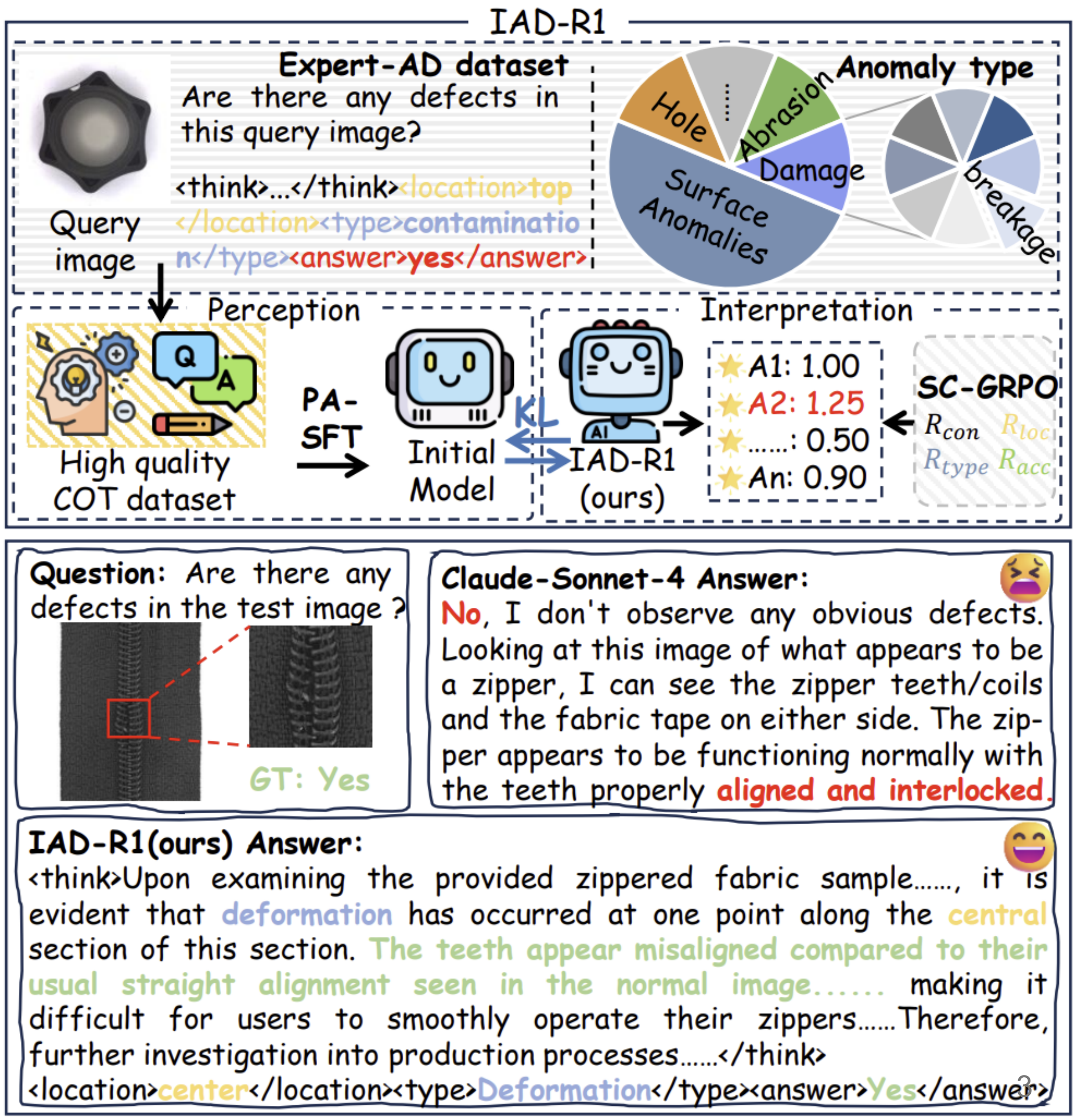
## Yes/No → Explainable Anomaly Detection

**Two-Stage Framework**

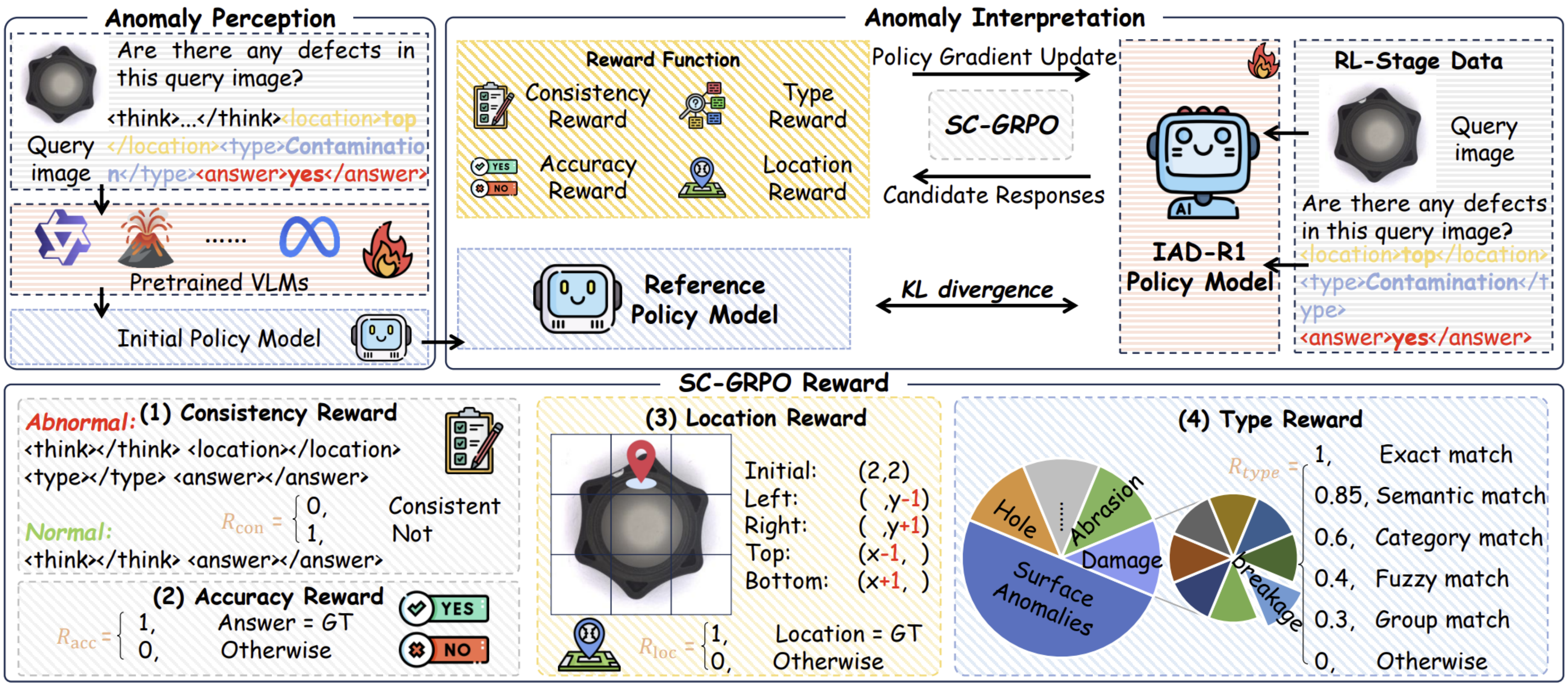
- Stage 1: PA-SFT (Supervised)
  - Expert-AD dataset (with CoT)
  - anomaly perception
- Stage 2: SC-GRPO (Reinforcement Learning)
  - Multi-dimensional reward: Accuracy, Location, Type, Consistency
  - 단순 정답이 아니라 논리적으로 맞는 reasoning 학습

**Example**

- 기존 모델: 오답 + 잘못된 reasoning
- IAD-R1: 정답 + 위치 + 유형 + 일관된 reasoning



# Methodology

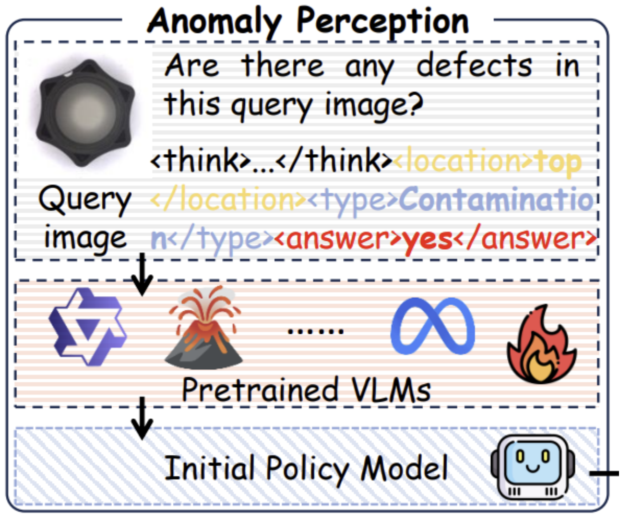


# Methodology

## PA-SFT (Perception Activation SFT)

### Training Data

- Dataset: Expert-AD
- Input format: (Image I, Prompt p, Output O)




### Output Structure

- Normal image: (Reasoning T + Answer A)
- Anomalous image: (Reasoning T + Location L + Type t + Answer A)

$$\mathcal{L}_{\text{PA-SFT}} = -\mathbb{E}_{(I,p,O)\sim\mathcal{D}_{\text{Expert-AD}}} \sum_{i=1}^{L} \log \pi_\theta(o_i | I, p, o_{<i})$$

### Key Idea

- CoT 기반 학습 → structured reasoning 생성 가능 & reasoning ↔ answer 일관성 확보

### Training Objective

- 다음 토큰 예측 (Language Modeling) & log-likelihood maximize

# Methodology

## SC-GRPO (Reinforcement Learning Stage)

### Input Model

- Initial policy: PA-SFT trained model

### PPO

- single reward + value network

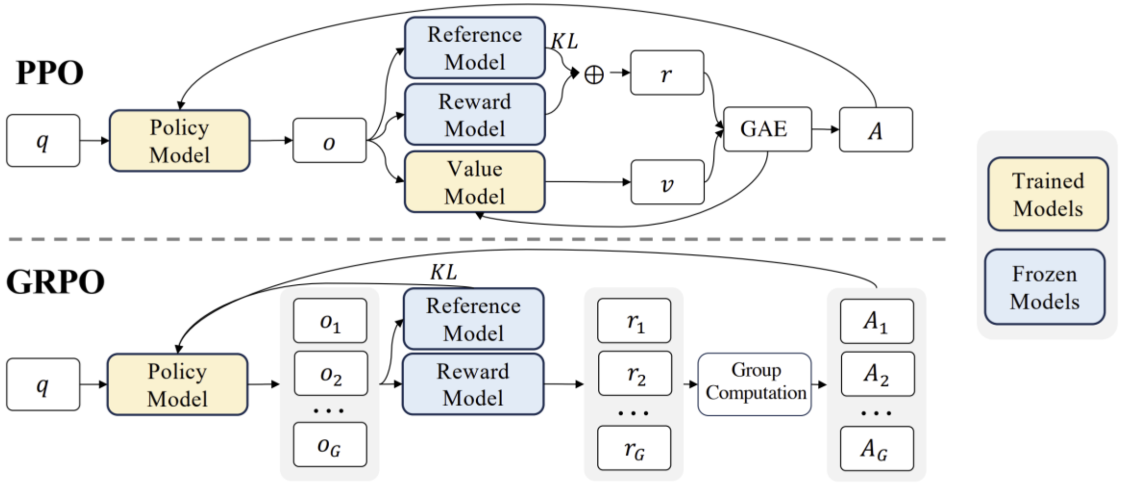
### GRPO

- relative reward + no value network

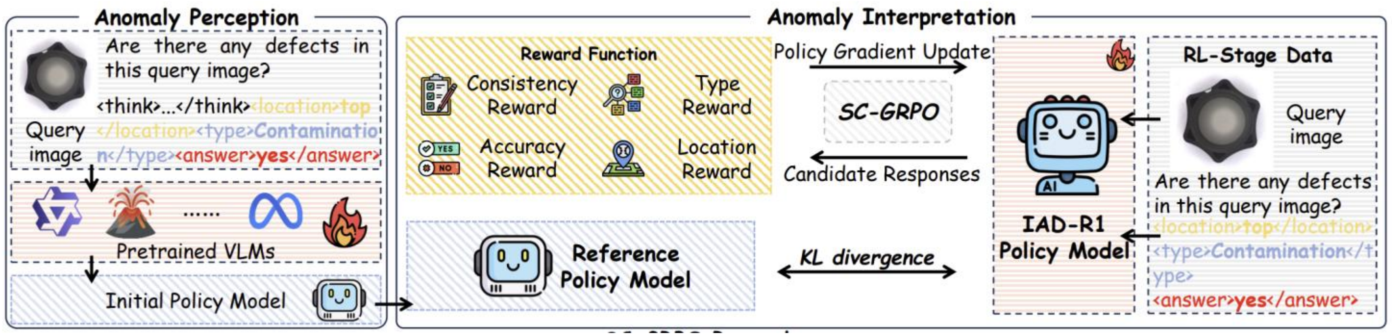
### SC-GRPO

- relative + multi-reward + structured reasoning

Many outputs from one query → select GRPO



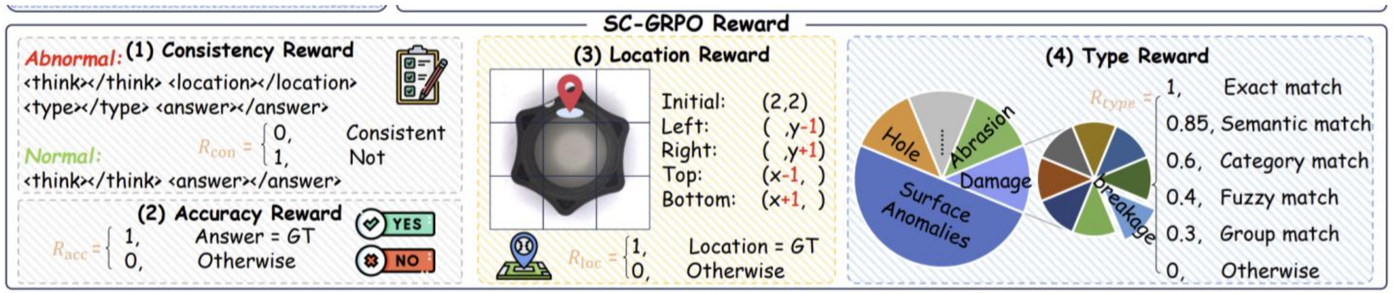

DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models




$$A_i = \frac{R_{\text{SC-GRPO}}(o_i) - \text{mean}\{R_{\text{SC-GRPO}}(o_j)\}_{j=1}^{G}}{\text{std}\{R_{\text{SC-GRPO}}(o_j)\}_{j=1}^{G}}.$$

# Methodology

## SC-GRPO (Reinforcement Learning Stage)




### Multi-dimensional Reward Design

- Consistency (R_con) → reasoning ↔ answer 일관성
- Accuracy (R_acc) → 정답 맞았는지
- Location (R_loc) → 위치 정확한지
- Type (R_type) → 결함 종류 맞는지

$$P_{normal} = \langle think \rangle ... \langle /think \rangle \langle answer \rangle ... \langle /answer \rangle.$$

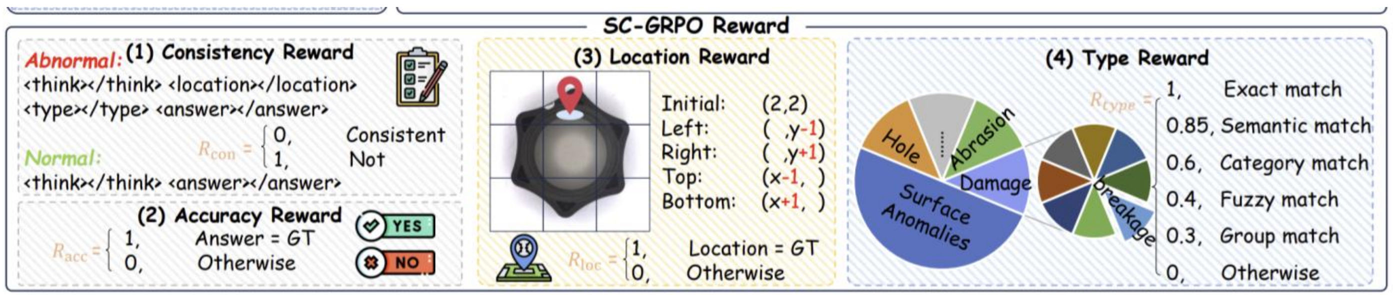
$$P_{abnormal} = \langle think \rangle ... \langle /think \rangle \langle location \rangle ... \langle /location \rangle \langle type \rangle ... \langle /type \rangle \langle answer \rangle ... \langle /answer \rangle.$$

### Consistency Reward

- 출력 구조를 강제해서 reasoning 일관성 확보함
- Format 일치하면 1, 불일치하면 0
- Reasoning과 answer간 inconsistency 방지함

$$R_{con}(o_i) = \begin{cases} 1, & \text{if Match}(o_i, P) \\ 0, & \text{otherwise} \end{cases}$$

# Methodology




## SC-GRPO (Reinforcement Learning Stage)

### Accuracy Reward

- Ground-truth랑 직접 비교하여 정확도 평가함
- Correct → 1, Wrong → 0

$$R_{acc}(a_{pred}, a_{gt}) = \begin{cases} 1, & \text{if } a_{pred} = a_{gt} \\ 0, & \text{otherwise} \end{cases}$$

### Location Reward

- 이상 위치 정확도 평가
- 텍스트 위치 → 3×3 grid mapping, GT와 비교
- 공간적 이해 학습

$$R_{loc}(l_{pred}, l_{gt}) = \begin{cases} 1, & \text{if } \Phi(l_{pred}) = \Phi(l_{gt}) \\ 0, & \text{otherwise} \end{cases}$$

### Type Reward

- 결함 유형 분류 정확도 평가
- 단순 string match가 아닌 다양한 표현 허용
- reward sparsity 완화

$$R_{type}(t_{pred}, t_{gt}) = \begin{cases} 1.0, & \text{if exact match} \\ 0.85, & \text{if semantic match} \\ 0.6, & \text{if category match} \\ 0.4, & \text{if fuzzy match} \\ 0.3, & \text{if group match} \\ 0, & \text{otherwise} \end{cases}$$

# Experiment

**Dataset**

- Industrial objects: MVTec-AD, VisA, MPDD
- Surface textures: DAGM, DTD, SDD

**Implementation**

- 다양한 VLM backbone 사용 (Qwen2-VL (2B), Qwen2.5-VL (3B, 7B), LLaVA 계열 (0.5B ~ 8B))

**Main Results**

- 최고 성능: 86.1% (IAD-R1, LLaVA-OneVision-SI-7B)
- 기존 SOTA 대비 +7.2% (open-source), +7.8% (GPT-4.1)
- 작은 모델도 강력함: 3B 모델 → 72B 모델보다 우수, 0.5B 모델 → 상용 모델보다 우수

**Insight**

단순 파라미터 증가보다 IAD-R1 학습 방식이 더 중요함

# Experiment

| Type | Model | Parameter | Industrial Workpieces | | | Surface Texture | | | Average |
|---|---|---|---|---|---|---|---|---|---|
| | | | MVTec | MPDD | VisA | DAGM | DTD | SDD | |
| Commercial | GPT-4o-mini | / | 71.3 | 67.9 | 65.1 | 72.6 | 79.5 | 66.6 | 70.5 |
| | GPT-4o | / | 69.6 | 60.3 | 63.5 | 63.0 | 69.9 | 65.7 | 65.3 |
| | GPT-4.1-nano | / | 74.7 | 61.7 | 60.5 | 62.4 | 78.3 | 50.0 | 64.6 |
| | GPT-4.1-mini | / | 74.0 | 69.8 | 63.4 | 70.7 | 82.1 | 72.1 | 72.0 |
| | GPT-4.1 | / | 81.9 | 66.7 | 69.1 | 81.8 | 90.1 | 79.9 | 78.3 |
| | Claude-Sonnet-4 | / | 67.6 | 65.9 | 63.5 | 69.2 | 88.4 | 81.7 | 72.7 |
| Open Source | LLaVA-OneVision-SI | 0.5B | 50.0 | 50.0 | 50.0 | 50.0 | 54.3 | 50.0 | 50.7 |
| | Anomaly-OV[CVPR 2025] | 0.5B | 50.0 | 50.0 | 50.0 | 50.0 | 53.8 | 50.0 | 50.6 |
| | Qwen2.5-VL-Instruct | 3B | 62.6 | 52.9 | 58.4 | 54.2 | 64.4 | 50.3 | 57.1 |
| | AnomalyR1[arxiv 2025] | 3B | 69.4 | 56.0 | 59.8 | 56.7 | 61.0 | 57.6 | 60.1 |
| | InternVL-2.5 | 4B | 56.6 | 59.1 | 53.7 | 57.7 | 81.3 | 64.1 | 62.1 |
| | Qwen2.5-VL-Instruct | 7B | 66.0 | 56.0 | 58.4 | 57.7 | 59.2 | 67.4 | 60.8 |
| | AnomalyGPT[AAAI 2024] | 7B | 46.6 | 54.2 | 57.3 | 49.6 | 64.1 | 49.5 | 53.6 |
| | LLaVA-OneVision-SI | 7B | <u>82.0</u> | 57.0 | 59.6 | 75.4 | 76.8 | 55.1 | 67.7 |
| | Anomaly-OV[CVPR 2025] | 7B | 74.3 | <u>70.3</u> | 74.3 | 77.5 | 90.7 | <u>88.7</u> | 78.9 |
| | LLaVA-1.5 | 13B | 61.4 | 61.4 | 67.2 | 50.4 | 75.9 | 50.0 | 61.1 |
| | LLaVA-1.6 | 34B | 53.7 | 50.0 | 53.9 | 50.0 | 52.7 | 50.0 | 51.7 |
| | Qwen2.5-VL-Instruct | 72B | 77.4 | 64.7 | 68.2 | 77.9 | 81.9 | 69.0 | 73.2 |
| IAD-R1 | IAD-R1(LLaVA-OneVision-SI) | 0.5B | 81.0 | 69.4 | 74.9 | <u>93.3</u> | <u>95.5</u> | 88.6 | <u>83.8</u> |
| | IAD-R1(Qwen2.5-VL-Instruct) | 3B | 77.6 | 59.2 | 69.8 | 86.0 | 89.1 | 84.5 | 77.7 |
| | IAD-R1(Qwen2.5-VL-Instruct) | 7B | 81.9 | 65.8 | <u>75.4</u> | 85.2 | 90.8 | 83.4 | 80.4 |
| | IAD-R1(LLaVA-OneVision-SI) | 7B | **86.7** | **70.9** | **78.0** | **94.8** | **96.2** | **90.1** | **86.1** |

# Experiment

## Ablation

### CoT Data Effect

- CoT 데이터 사용 시 모든 모델에서 성능 향상됨
- 단순 QA 학습 → shallow learning / overfitting
- CoT 학습 → 깊은 anomaly reasoning 학습 가능

### Reward Contribution

- 각 reward 모두 성능에 기여함
- 전체 조합이 가장 좋음

### Grid Size (Location)

- 3×3 grid이 best
- 너무 작으면 정보 부족하고 너무 크면 noise 증가함

| Model | Data | 0-shot | 1-shot | Average |
|---|---|---|---|---|
| LLaVA-OneVision-SI (0.5B) | Base | 50.7 | 50.0 | 50.4 |
| | Original | 50.9 | 50.0 | 50.5 |
| | Expert-AD | **82.7** | **69.5** | **76.1** |
| Qwen2.5-VL-Instruct (3B) | Base | 57.1 | 60.3 | 58.7 |
| | Original | 65.8 | 59.5 | 62.7 |
| | Expert-AD | **73.1** | **73.5** | **73.3** |
| LLaVA-OneVision-SI (7B) | Base | 67.7 | 60.6 | 64.2 |
| | Original | 68.4 | 62.3 | 65.4 |
| | Expert-AD | **85.9** | **74.9** | **80.4** |

Table 2: Results of different data preparation in PA-SFT.

| $R_{con}$ | $R_{loc}$ | $R_{type}$ | MVTec | VisA | DAGM | MPDD | SDD | DTD |
|---|---|---|---|---|---|---|---|---|
| | | | 72.7 | 59.0 | 72.5 | 55.5 | 79.3 | 81.0 |
| | ✓ | | 76.3 | 66.3 | 83.9 | 57.9 | 77.4 | 88.4 |
| | | ✓ | 77.4 | 67.0 | 84.8 | 57.9 | 81.6 | 88.1 |
| ✓ | ✓ | | 75.4 | 67.0 | **85.8** | 58.6 | 79.5 | **90.0** |
| ✓ | | ✓ | 76.7 | 67.5 | 85.1 | 56.5 | 80.8 | 89.1 |
| ✓ | ✓ | ✓ | **77.6** | **69.8** | 85.2 | **59.2** | **83.4** | 89.1 |

Table 5: Ablation of reward function contributions.

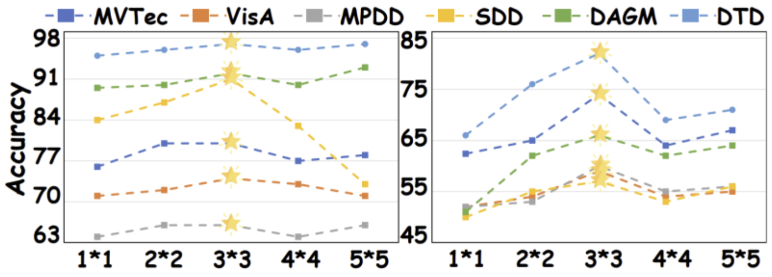



Figure 3: Ablation on grid size.

# Conclusion

**Overall Perspective**

- IAD-R1은 VLM 기반 이상탐지에서의 reasoning 부족과 일관성 문제를 해결하기 위해 제안됨
- Expert-AD (CoT 데이터)와 PA-SFT + SC-GRPO를 통해 reasoning 기반 학습 수행
- 단순 탐지를 넘어 Anomaly Perception → Interpretation으로 확장
- 다양한 벤치마크에서 성능 향상 및 높은 효율성 확인

**RL perspective**

1. Value-free RL (GRPO) → Value estimation 없이도 안정적 학습 가능
2. Reward Design의 중요성 → multi-dimensional reward
3. Structured Output + RL (output format을 강제하고 reward로 반영)→ RL로 reasoning 구조까지 학습 가능

# Q & A